Report 1: Initial Progress Report

Weighted-RVB Initialization and Heisenberg-HVA Refinement
for a 19-Site Kagome Heisenberg Benchmark

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QIntern 2026 Project qi26_09
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July 2026

1. Project Idea

This is an initial progress report, not a final research manuscript. The goal is to test a potential no-calibration route for preparing low-energy states of a 19-site Kagome antiferromagnetic Heisenberg benchmark. The main idea is to improve the variational state structure rather than modify the target Hamiltonian.

Potential solution. Start from a physically motivated signed weighted resonating-valence-bond (RVB) state, then refine it using shallow edge-colored Heisenberg Hamiltonian variational ansatz (HVA) layers. The target Hamiltonian remains fixed:

$$H = \sum_{\langle i,j \rangle} (X_i X_j + Y_i Y_j + Z_i Z_j).$$

The exact fixed-sector benchmark energy for the 19-site patch is

$$E_0 = -29.146168109350.$$

The current preliminary no-calibration result reaches

$$E = -29.037601351802, \qquad F = 0.982518440088$$

at HVA depth $p = 4$.

Figure 1 summarizes the project as two connected branches. The exact-reference branch diagonalizes the fixed 19-site Hamiltonian to obtain E_0 and $|\psi_0\rangle$. The ansatz branch builds a weighted-RVB state, refines it with Heisenberg-HVA layers, and then compares the resulting trial state to the exact reference using energy, fidelity, and bond correlations. The dashed calibration-scan box is only future design guidance.

Repository / current work link.
https://github.com/adoolelomani2026/qintern-kagome-hva

One-sentence status. The current evidence is a finite-size proof-of-concept that weighted-RVB structure plus shallow Heisenberg-level refinement can move a dimer/RVB state close to the exact 19-site ground state, but it does not yet prove scalable or hardware-ready quantum spin liquid preparation.

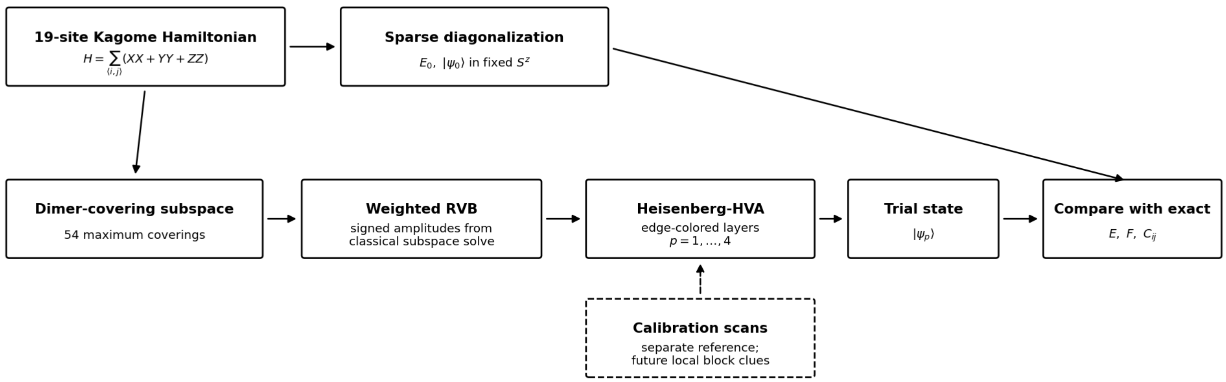

Figure 1: Project schematic. The target Hamiltonian is fixed. The upper branch produces the exact reference state, while the lower branch produces the variational trial state through weighted-RVB initialization and Heisenberg-HVA refinement. Calibration scans are shown with a dashed box because they are future design guidance, not part of the present no-calibration claim.

2. Background and Benchmark

The Kagome antiferromagnetic Heisenberg model is a standard frustrated-magnetism benchmark because its geometry makes simple classical ordering difficult. In the thermodynamic limit, this model is closely connected to quantum spin liquid physics, with DMRG studies supporting a spin-liquid ground-state picture [1, 2]. Here the project uses a smaller 19-site open-boundary patch because it is still large enough to show nontrivial frustration while remaining small enough for exact fixed-S^z diagonalization.

The active benchmark is the 19-site graph and bond list confirmed from the project provenance material. The calculation is performed in the fixed sector with $n_\downarrow = 9$, whose Hilbert-space dimension is

$$\dim \mathcal{H}_{n_\downarrow=9} = \binom{19}{9} = 92{,}378.$$

This is large enough that naive full statevector work is inconvenient, but small enough for sparse diagonalization and optimized fixed-sector NumPy kernels.

Why this benchmark is useful. A static dimer product is a natural starting point, but it is too rigid: it locks singlet weight onto specific bonds. A more flexible RVB state allows many dimer coverings to interfere with signs and amplitudes. The HVA then adds circuit-compatible two-qubit Heisenberg evolution gates:

$$U_{ij}(\theta) = \exp[-i\theta(X_iX_j + Y_iY_j + Z_iZ_j)].$$

Role of calibrated-Hamiltonian work. The calibrated-Hamiltonian direction discussed in the project is treated as methodology context and as a source of ansatz-design clues [5]. In this initial no-calibration route, the Hamiltonian used for evaluation is not changed. Instead, future work can use the information from sensitive triangles or bonds to design better circuit blocks.

Figure 2 visualizes why the weighted-RVB initializer is a better starting point than a single frozen dimer covering: antiferromagnetic bond weight is distributed across the graph instead of being locked onto only a few perfect singlets.

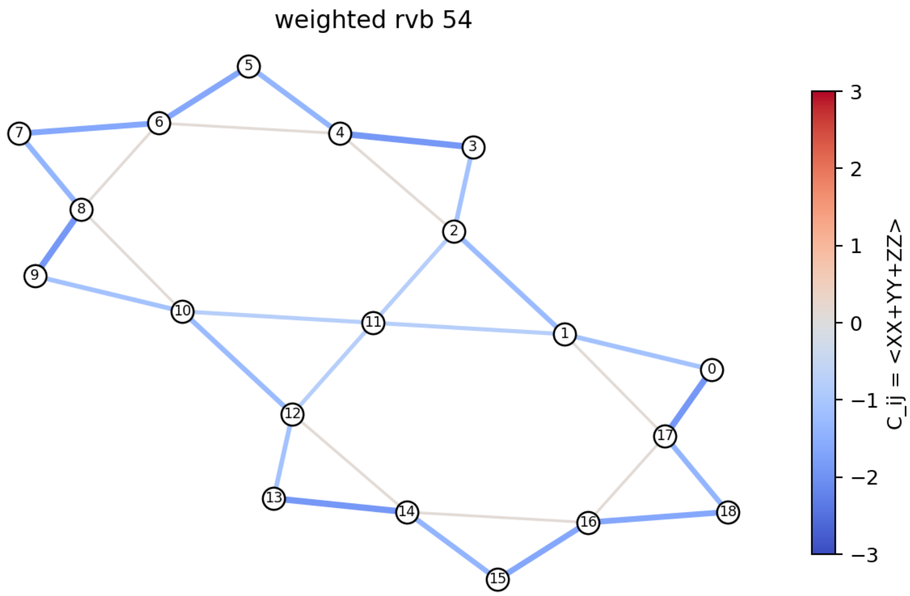

Figure 2: Bond-correlation map for the weighted-RVB initializer. This finite-patch diagnostic shows that antiferromagnetic correlations are distributed over the graph rather than frozen onto one dimer covering.

3. Current Method

The current workflow has two main stages.

Stage 1: weighted RVB initialization. The code enumerates 54 maximum dimer coverings on the 19-site graph. Instead of taking an equal positive superposition, it solves a classical generalized eigenproblem in the dimer-covering subspace. This produces a signed weighted-RVB state. The equal RVB state is not enough; the signs and amplitudes are important.

Stage 2: edge-colored Heisenberg-HVA refinement. The 30 graph bonds are partitioned into four disjoint edge-color groups. Within each group, gates act on disjoint bonds and can be applied in parallel. This follows the broader variational quantum eigensolver and hardware-efficient ansatz philosophy, but uses Heisenberg interaction gates matched to the model [3, 4]. A depth-p ansatz therefore uses only $4p$ HVA parameters:

$$U_p = \prod_{\ell=1}^{p} \prod_{c=1}^{4} \prod_{(i,j)\in E_c} \exp[-i\theta_{\ell,c}(X_iX_j + Y_iY_j + Z_iZ_j)].$$

For the current best no-calibration run, $p = 4$, so the HVA refinement uses 16 parameters.

Why this is a reasonable first route. The ansatz is physically matched to the Hamiltonian because every two-qubit gate is a Heisenberg interaction. It also avoids immediately relying on Hamiltonian calibration. That keeps the scientific question clean: how far can a better initial state plus physically structured circuit layers go against the original target Hamiltonian?

Figure 3 is a compact dashboard of the current proof-of-concept. It is not a separate calculation; it summarizes the same exact-benchmark comparison reported in the tables and CSV files.

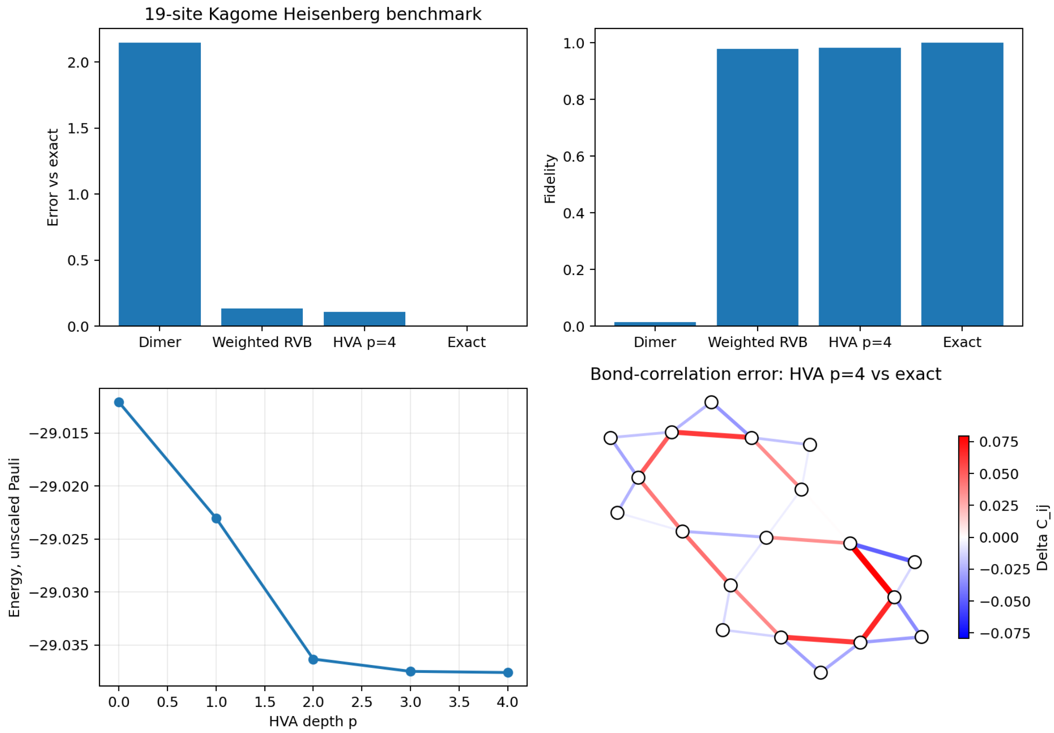

Figure 3: Compact summary of the current finite-size proof-of-concept. Weighted-RVB gives the main jump in quality; shallow Heisenberg-HVA layers add no-calibration refinement.

4. Preliminary Results

The table below summarizes the current main result in the unscaled Pauli convention. Energy and fidelity are always measured against the same fixed target Hamiltonian and the exact 19-site ground state.

State	Energy	Error vs exact	Fidelity
Static dimer	-27.00000000	2.14616811	0.014241
Equal RVB-54	-26.39781307	2.74835504	0.004371
Weighted RVB-54	-29.012072100265	0.13409601	0.978535993636
Weighted RVB + HVA $p = 2$	-29.03633182	0.10983629	0.982446
Weighted RVB + HVA $p = 4$	-29.037601351802	0.10856676	0.982518440088
Exact	-29.146168109350	0.00000000	1.000000

The static dimer state is far from the exact state. The equal RVB state is also poor, which shows that merely superposing dimer coverings is not enough. The signed weighted-RVB state gives the largest improvement, reaching fidelity about 0.9785. The Heisenberg-HVA refinement then gives a smaller but useful improvement, reaching fidelity about 0.9825 at $p = 4$.

Figure 4 shows the depth dependence directly. The energy improves sharply through $p = 2$ and then begins to saturate, while the fidelity remains high and changes only modestly beyond the weighted-RVB starting point.

Current interpretation. The result is promising as a finite-size baseline. It suggests that improving the state structure matters more than simply adding arbitrary circuit depth. However, the remaining no-calibration energy gap is still about 0.1086, so this is not yet a finished solution.

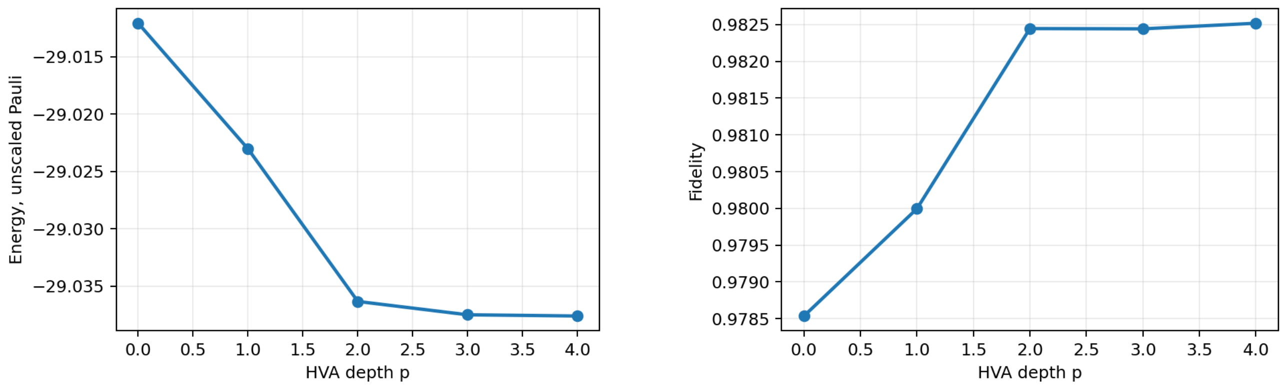

Figure 4: Energy and fidelity versus HVA depth after weighted-RVB initialization. Energy improves most through $p = 2$ and then begins to saturate, while fidelity remains high.

5. Next Step and Limitations

The most useful next step is not a blind depth sweep. The controlled next experiment is calibration-informed ansatz design: use the sensitive local structures identified by calibration studies as extra circuit blocks, but keep the target Hamiltonian unchanged. For example, add a local Heisenberg block on triangle $(6, 7, 8)$ or bond $(14, 16)$:

$$U_\triangle(\alpha) = \prod_{(i,j)\in\triangle} \exp[-i\alpha(X_iX_j + Y_iY_j + Z_iZ_j)].$$

This tests whether calibration information can guide circuit expressibility without becoming Hamiltonian calibration [5]. Figure 5 is the motivation for this idea, but it needs to be read carefully. The horizontal axis is the energy error relative to the exact 19-site target, $|E - E_{\text{exact}}|$, so points farther left are better. The vertical axis is the fidelity with the exact state, so points higher up are better. The ideal location is therefore the upper-left corner: nearly zero energy error and fidelity nearly one.

The colored calibration points are not HVA circuits. Each one is the exact ground state of a modified Hamiltonian in which selected bonds, edge-color groups, or triangles are given a different coupling J', and that exact state is then compared back to the original uniform-Hamiltonian target. The red star is different: it is the present no-calibration weighted-RVB plus HVA $p = 4$ state. Its error is about 0.1086 and its fidelity is about 0.9825, so it is good compared with the static dimer baseline but still noticeably to the right of the best calibrated reference states.

Plot element	Meaning
Green, blue, orange points	Exact ground states of modified Hamiltonians, used as reference diagnostics.
Red star	Current no-calibration weighted-RVB + HVA $p = 4$ state.
Upper-left direction	Better agreement with the exact 19-site target: lower energy error and higher fidelity.

The conclusion is therefore not that the no-calibration HVA already beats calibrated Hamiltonian references. It does not. The conclusion is that the calibration scans identify local structures, especially triangles, that can be converted into extra HVA circuit blocks while still evaluating everything against the original unmodified Hamiltonian.

Limitations at this stage.

- The result is finite-size only and uses a 19-site open-boundary patch.
- The weighted-RVB initializer is obtained classically and is not yet a hardware-preparable state-preparation circuit.

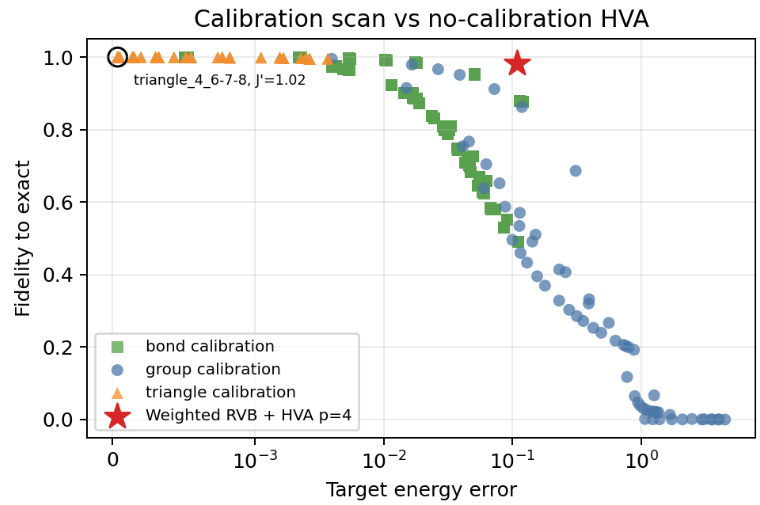

Figure 5: Calibration scan landscape compared with the no-calibration HVA baseline. Left is lower target-energy error and up is higher exact-state fidelity, so the upper-left corner is best. The green, blue, and orange points are exact reference states from modified Hamiltonians; the red star is the present no-calibration weighted-RVB + HVA $p = 4$ result. The figure is used to choose future local circuit blocks, not to claim that the present HVA already outperforms calibration.

- The HVA improvement is real but modest through $p = 4$.
- No statistical robustness campaign, noise model, or hardware execution is claimed in this initial report.
- This report should be read as a progress report and potential solution direction, not as a final proof.

References

[1] S. Yan, D. A. Huse, and S. R. White, "Spin-liquid ground state of the $S = 1/2$ kagome Heisenberg antiferromagnet," *Science* **332**, 1173–1176 (2011).

[2] S. Depenbrock, I. P. McCulloch, and U. Schollwöck, "Nature of the spin liquid ground state of the $S = 1/2$ Heisenberg model on the kagome lattice," *Physical Review Letters* **109**, 067201 (2012).

[3] A. Peruzzo *et al.*, "A variational eigenvalue solver on a photonic quantum processor," *Nature Communications* **5**, 4213 (2014).

[4] A. Kandala, A. Mezzacapo, K. Temme, M. Takita, M. Brink, J. M. Chow, and J. M. Gambetta, "Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets," *Nature* **549**, 242–246 (2017).

[5] M. Ahsan, "Utility-scale Experimental Quantum Computation with Hardware Efficient Ansätze and Calibrated Hamiltonian," *ACM Transactions on Quantum Computing* (2026), doi:10.1145/3815786.